

Functions and transformations - Set 1

TMUA practice questions by topic. Specification MM1.7, MM8.2, MM8.3, MM8.4. 12 questions.

Each question has one correct answer. No calculator. Answers and worked solutions are in the separate solutions PDF.

- 1 The graph of $y = x^2$ is translated by 3 units in the positive x -direction and then stretched parallel to the y -axis with scale factor 2. What is the equation of the resulting graph?
- A** $y = 2(x - 3)^2$
- B** $y = (x - 3)^2 + 2$
- C** $y = 2(x + 3)^2$
- D** $y = 2x^2 - 3$
- E** $y = (2x - 3)^2$
- F** $y = \frac{1}{2}(x - 3)^2$
- 2 $f(x) = x^2 - 4x$. The graph of $y = f(x + 2)$ has its minimum point at
- A** $(0, -4)$
- B** $(2, -4)$
- C** $(4, -4)$
- D** $(2, 0)$
- E** $(-2, -4)$
- F** $(0, 0)$
- 3 Let $f(x) = 2x + 1$ and $g(x) = x^2 - 3$. Find $f(g(2)) - g(f(2))$.
- A** -3
- B** -17
- C** -19
- D** 3
- E** 25
- F** 19

- 4 The graph of $y = f(x)$ passes through (4, 6). Which point must lie on the graph of $y = 3f(2x) - 1$?
- A (2, 11)
B (2, 17)
C (8, 17)
D (8, 18)
E (4, 17)
F (2, 18)
- 5 The curve $y = x^2 - 6x + 5$ is reflected in the y-axis. Find the equation of the reflected curve.
- A $y = x^2 + 6x + 5$
B $y = x^2 + 6x - 5$
C $y = -x^2 - 6x + 5$
D $y = -x^2 + 6x - 5$
E $y = x^2 - 6x - 5$
- 6 For which values of x is $\sqrt{x^2} = -x$?
- A $x = 0$ only
B $x \geq 0$
C $x \leq 0$
D $x < 0$ only
E all real x
F no real x
- 7 The graph of $y = \sin x$ is transformed to $y = 2\sin(3x) + 1$. What is the maximum value of the new function, and what is its period (in degrees)?
- A maximum 1, period 120°
B maximum 3, period 360°
C maximum 2, period 120°
D maximum 3, period 120°
E maximum 3, period 1080°
F maximum 2, period 180°

- 8** The graph of $y = f(x)$ is transformed by the sequence: reflect in the x -axis, then translate 2 units in the positive y -direction, then reflect in the y -axis. The resulting graph has equation
- A** $y = f(-x) - 2$
- B** $y = 2 - f(x)$
- C** $y = -f(-x + 2)$
- D** $y = 2 - f(-x)$
- E** $y = -f(x - 2)$
- F** $y = -f(-x) - 2$
- 9** The graph of $y = a(x + b)^2 + c$ has vertex $(-2, 5)$ and passes through $(0, -3)$. Find a .
- A** -2
- B** 2
- C** -4
- D** -1
- E** 1
- F** $-\frac{1}{2}$
- 10** The graph of $y = f(x)$ has a maximum point at $(3, 8)$. The graph of $y = f(x - a) + b$ has its maximum at $(1, 2)$. Find $a + b$.
- A** 8
- B** -2
- C** -4
- D** -8
- E** 4
- F** 0

- 11** Which of the following is the equation of the graph obtained by stretching $y = x^3 - 3x$ parallel to the x -axis with scale factor $\frac{1}{2}$?
- A** $y = \frac{1}{2}x^3 - \frac{3}{2}x$
- B** $y = 8x^3 - 3x$
- C** $y = 2x^3 - 6x$
- D** $y = \frac{x^3}{8} - \frac{3x}{2}$
- E** $y = 8x^3 - 6x$
- 12** The line $y = mx + c$ passes through $(2, 7)$. If m is increased by 1 and c is decreased by 2, the new line passes through $(2, k)$. Find k .
- A** 6
- B** 7
- C** 5
- D** 9
- E** 8
- F** 10